U-space USSP-USSP Exchanges for UAS Flight Authorisation SWIM Service Description Document

General Service Description	
Service description identification **SWIM-SERV-030**	**Title:** U-space USSP-USSP Exchanges for UAS Flight Authorisation SWIM Service Description **Edition:** V2.0.1 **Reference Date: 2026-11-05**
Service identification **SWIM-SERV-040**	**Name:** U-space USSP-USSP Exchanges for UAS Flight Authorisation Service **Version:** V2.0
Service abstract **SWIM-SERV-050**	The UAS Flight Authorisation service is one of the four mandatory services under the Commission Implementing Regulation (EU) 2021/664. The UAS flight authorisation service provides authorisations to UAS operators for each individual flight based on other notified flight requests that may conflict with other unmanned operations within the same U-space airspace. It is a strategic deconfliction tool. This service implementation is based on the ASTM F3548-21 standard, identified as an applicable standard in U-space accepted means of compliance and guidance material.
Service Provider and Contact Information	
Service Provider **SWIM-SERV-060**	**Name:** U-space Service Provider (see Operations Committee list for participating providers) **Description of organization:** U-space Service Provider refers to an organisation that is certified by a competent authority to provide U-space services in the U-space airspace (Ref: GM1a Article 7 U-space service providers)
Service Characteristics	
Geographical extent of information **SWIM-SERV-090**	Geographical coverage of exchanged information service payload is available through the DSS under ASTM Section A2.

<table>
<tr><td>Service categories

SWIM-SERV-100</td><td>Service category: Flight information</td></tr>
<tr><td>Service lifecycle information

SWIM-SERV-110</td><td>Lifecycle stage and if applicable qualifying dates: As published by the Operations Committee.</td></tr>
<tr><td>Service standard reference

SWIM-SERV-120</td><td>Service standard reference: Standard Specification for UAS Traffic Management (UTM) UAS Service Supplier (USS) Interoperability (ASTM F3548-21)

Implemented options of the service standard:

● Per U-space regulation Article 10, conflicts are not permitted within the same priority level
● Requirements in Section A2.4 regarding down USS not used
● Per Article 10(8), U-space service providers shall give priority to UAS conducting special operations using the following priority levels

<table>
<tr><td>Operation Type</td><td>ASTM F3548-21 Priority Level</td></tr>
<tr><td>SPECIAL</td><td>100</td></tr>
<tr><td>NORMAL</td><td>0</td></tr>
</table>

Discovery and synchronization service configuration process - as defined by InterUSS DSS Pooling instructions on joining an existing pool with new instance.

DSS deployment requirements (minimum)
Each USSP joining the DSS pool shall deploy its DSS instance with the following configuration:

● DSS version: v0.23.0
● Datastore: CockroachDB
● Datastore version: v24.1.3
● Number of nodes: 3
● Configurable flags: None

Each DSS node shall meet the following minimum specification:

● Dedicated vCPUs: 4
● Memory (RAM): 16 GB
● Storage: 20 GB minimum
● Network latency between DSS nodes in the pool should be</td></tr>
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<table>
<tr><td></td><td>kept below 20 ms round-trip 95% of the time

DSS clusters

<table><tr><th>Environment</th><th>Cluster name</th></tr><tr><td>Test (Gate 2)</td><td>uspace-zurich-test</td></tr><tr><td>Production (Gate 3)</td><td>uspace-zurich-ops</td></tr></table>
Note: These clusters may be shared with other U-space services that rely on the DSS. A separate cluster is not required for the UAS Flight Authorisation Service.</td></tr>
<tr><td colspan="2">High-level Description of Service Offer</td></tr>
<tr><td>Operational environment

SWIM-SERV-130</td><td>Refer to the following documents for operational needs addressed by the service and the capabilities offered by the service.

● Commission Implementing Regulation (EU) 2021/664 - Preamble
● ASTM F3548-21 Section A2 capabilities include:
- USS-DSS Interfaces
- Other DSS Interfaces
- USS-USS Interfaces
- DSS-DSS Synchronization</td></tr>
<tr><td>Service functions

SWIM-SERV-140</td><td>Refer to the following documents for functions offered by the service and their real world effects.

● Commission Implementing Regulation (EU) 2021/664 Article 10
● Article 10 Accepted Means of Compliance and Guidance Material to Regulation (EU) 2021/664 — Issue 1
● ASTM F3548-21 Section A2 functions are provided for the following capabilities:
- USS-DSS Interfaces
- Other DSS Interfaces
- USS-USS Interfaces
- DSS-DSS Synchronization</td></tr>
<tr><td colspan="2">Limitations and Constraints on Using the Service</td></tr>
<tr><td>Service access and use conditions

SWIM-SERV-150</td><td>Refer to the EU USSP U-space Data Sharing and Governance Agreement for information on legal constraints, service policies, and service consumption constraints

Service operational policies:
With reference to Regulation (EU) 2021/664:
*
● USSPs shall use 4D volumes exchanged as a part of this</td></tr>
</table>

	service to detect conflicts among flight authorisation requests. The USSP-Operator agreement will ensure that the 4D volumes exchanged among USSPs consider U-space requirements such as inclusion of deviation thresholds and contingency/emergency measures as applicable ● For the purposes of V1, deviation thresholds are set to be 0 ● * A variant 1 (RFC 4122), Version 4 UUID shall be used for all shared flight authorisations.
Security constraints **SWIM-SERV-160**	. For field testing (U-space v2), security compliance shall be demonstrated either by: ○ holding a valid USSP certification in accordance with Regulation (EU) 2021/664, or ○ holding an ISO/IEC 27001 certification and being in the process of obtaining USSP certification, or ○ complying to comparable security industry standards (such as IEC 62443) and being in the process of obtaining USSP certification. This recognition is a transitional measure limited to field testing and is not applicable to full U-space deployment. ASTM F3548-21 Sections 5.2 and A2 (relevant requirements are listed below): * GEN0005 * DSS0200 * DSS0205 *Per ASTM F3548-21, Appendix X1 - Reference architecture for interoperability security controls, The X1.1 Base Deployment: Access Tokens with Audience Claims is implemented. Authentication and Authorization for communication between USS-USS and USS-DSS shall be facilitated by a common interoperability token exchanger service supporting OAuth 2.0.*
Quality Aspects	
Quality of service **SWIM-SERV-180**	Refer to the following documents for quality requirements. ● EU USSP U-space Data Sharing and Governance Agreement - Annex XX Service Level Agreement for information on quality of service
Service validation information **SWIM-SERV-200**	Service will be verified using the InterUSS Automated Test Baseline prior to software deployment using test baseline identifier TB-ff672da.
Behavior of the Service	

Application message exchange pattern **SWIM-SERV-210**	Refer to the following sections in ASTM F3548-21 for information about the application message exchange pattern(s) used by the service. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description
Service behavior **SWIM-SERV-220**	Refer to the attached Compliance Matrix, *U-space Article 10 UAS Flight Authorisation Compliance Matrix V2.0* for service performance and verification requirements
Service Implementation and Structural Details	
Service interfaces **SWIM-SERV-240**	Refer to the following sections in ASTM F3548-21 for information about service interfaces. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description
Service interface protocols and data format **SWIM-SERV-260**	Refer to the following sections in ASTM F3548-21 for information about service interface protocols and data format. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description
Service operations **SWIM-SERV-270**	Refer to the following sections in ASTM F3548-21 for information about service operations. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description
Service messages **SWIM-SERV-280**	Refer to the following sections in ASTM F3548-21 for information about messages exchanged by the service. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description
Information Aspects of the Service	
Information definition (minimum) **SWIM-SERV-290**	Refer to the following sections in ASTM F3548-21 for information about the exchanged information service payload. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description
Resources	
Machine-readable service interface definition **SWIM-SERV-320**	Refer to the following sections in ASTM F3548-21 for information about the exchanged information service payload. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description

Abbreviations and acronyms **SWIM-SERV-350**	Refer to the following sections in ASTM F3548-21 for abbreviations and acronyms: * 3 - Terminology * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description